

SID DIAMOND

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Personal statement

Master's student in Physics at Imperial College London researching directed acyclic graphs (DAGs) within the Theoretical Physics Group. Currently supplementing this thesis with courses in information theory and general relativity, alongside research into how benchmark datasets in supervised machine learning can create bias in hate speech detection systems. My undergraduate background includes mathematical analysis, foundations of quantum mechanics, and the highest mark in the cohort in Communicating Physics for a policy-focused educational report. My bachelor's dissertation was on uncertainty quantification in transformer algorithms using Fisher information, asymptotic normality conditions and Monte Carlo dropout. Last summer I undertook an internship as lead developer for a minimum viable product multi-agent planning application at a software company. The previous summer I conducted an independent research project under the guidance of the Imperial High Performance Computing (HPC) research group, using transformer algorithms to estimate a per-document probability of actionable insight for climate policy documents.

Education

Imperial College London: Integrated MSci in Physics 2022- Present

- Year One: 2:1 | Year Two: 2:1 | Year Three: 2:1 | Masters Year: On track for 1st class
- First-class results in all programming. Object-Oriented Programming (98.7%)
- Master's thesis on modelling innovation networks and applying citation network analysis to the artificial networks (1st class for literature review).
- Statistics (72.9%)
- Problem solving (79.3%)
- BSc Thesis in Uncertainty in Large Language models (69.50%)
- Communicating Physics (84.1%)
- Currently also studying Information theory, a Sociology essay elective (on benchmark datasets), and General Relativity

Stoke Newington School and Sixth Form 2014 -2021

- A-level Subjects: Further Maths (A*), Maths (A*), Physics (A*), Chemistry (A*), EPQ (A*)
- Chemistry Olympiad Silver 2021
- Kelvin Prize winner 2020 (*Peterhouse College Cambridge University Science Competition*)
- GCSE Subjects: 3 grade 9s, 3 grade 8s, 3 grade 7s

Research Projects

Citation Networks & Innovation | *Master's Thesis* (Summer 2025–Spring 2026)

Developed both a new analytical probability distribution for the in-degree of nodes in finite-size DAGs whose growth is governed by cumulative advantage (high in-degree nodes are more likely to attract more in-degrees) and homophily (nodes with similar characteristics being more likely

to connect), and a new Kolmogorov–Smirnov and parametric bootstrap based goodness-of-fit algorithm to account for the non-independent, finite-size, discrete data subsets arising in DAG models of real-world networks.

- First-class mark achieved for all so-far marked aspects: literature review, scientific poster and viva.
- Working under the supervision of Associate Professor Tim Evans, Theoretical Physics Group, Centre for Complexity Science.
- The paper, not yet marked although my supervisor has reviewed and confirmed my results are extensions to current literature, is in the process of being shared with visiting academics given these novel contributions.

BERT Economic Policy Inference Software | *Imperial HPC + BSc Thesis* Summer–Winter 2024

- Developed a Bidirectional Encoder Representations from Transformers (BERT)-based NLP tool to extract policy topics, producing, for example, the per document probability of actionable insight for economic climate policy documents, achieving near-industry accuracy (ClimateBERT dataset)
- Developed into undergraduate thesis on uncertainty quantification in large language models; investigated inverse Fisher Information based variance estimates, implemented Monte Carlo dropout for transformer uncertainty, and analysed information loss from UMAP dimensionality reduction
- Worked independently over the summer under guidance from the HPC Research Group, before continuing under the supervision of Dr Aidan Crilly, Physics, Imperial College Research Fellow

A-Level Physics Attainment Gap Report | *Communicating Physics* Spring 2023

- Authored evidence-based report on state vs. private school preparation, integrating quantitative attainment data with qualitative classroom observations
- Awarded cohort-best mark; demonstrated ability to write clear, transformative insights for policymakers

Employment

Product Engineer Intern at Mindlace Ltd Summer 2025

- Led development of an AI agent-driven planning application MVP. Built multi-agent architectures that autonomously coordinated complex booking workflows
- Used modern AI agent software (n8n, Supabase, OpenRouter, Pinceone) to create the entirely unsupervised booking of special requirement meeting itineraries requested by users

Science, English, and Maths Teacher and Tutor 2021 – 2024

- Tailored explanations to meet diverse learner needs, effectively supported a range of abilities and neurodivergence in pupils
- Demonstrated leadership in 4-month weekly placement independently teaching GCSE and A-level Physics to classrooms of 30 students at Brentside High School, Ealing

Landways Management Summer 2021

- Evidenced reconfiguration of weak network strengths in a football stadium and subsequent maintenance of the improved systems
- Installation of hardware based on technical understanding

Ove Arup Engineering Internship Spring 2019

- Implemented creative solutions to reduce project timelines on internal software
- Punctual and professional temperament

Software Experience

- Python. Relevant libraries: PyTorch, SciPy, NumPy, Pandas, Matplotlib, Plotly
- NetLogo
- Minimal Viable Product Software Engineering tools: n8n, Shadcn/ui, Supabase, OpenRouter, Pinecone, Firecrawl, Tavily

Hobbies & Interests

Football & Running

- Imperial Men's Football 1st and 2nd teams, previously acting captain of 2nd team
- Enjoy running. Previously competitively at nation level however now recreationally

Journalism & Music

- Comment Piece writer for, Felix, the Imperial College student newspaper. Writing on: Artificial Intelligence
- To wind down, I like to play the guitar